DO NOW — RECALL

Lesson 067 · Divide and Conquer: Splitting the Search

Name: _____ Class: _____ Date: _____

1. Define decomposition.

2. Which input assumption does binary search require?

3. How does Math.Min(a,b) behave?

4. Binary search [2,4,6,8,10,12,14] for 12: show low/high and midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Breaking a complex problem into smaller, manageable parts. The parts may depend on one another.

2. Model answer

Sorted elements.

3. Model answer

Returns the smaller argument.

4. Model answer

low0 high6 mid3=8, then low4 high6 mid5=12; found index5.

By the end, you can…

• Divide and conquer breaks a problem into smaller instances, solves them, and (where relevant) combines results.

• Binary search repeatedly discards half of a SORTED array. It is an example of reducing to one subproblem rather than solving both.

• Merge sort divides into halves and combines them in sorted order. Dividing alone is not enough; correctness depends on valid preconditions and combining.

Vocabulary — know these terms

divide and conquer — Solve smaller subproblems and combine.

precondition — Requirement for correctness.

binary search — Halving search on sorted data.

merge — Combine sorted partial results.

NEW LEARNING

Read, annotate and explain

Define and predict

Divide and conquer breaks a problem into smaller instances, solves them, and (where relevant) combines results.

Explain

Binary search repeatedly discards half of a SORTED array. It is an example of reducing to one subproblem rather than solving both.

Evaluate

Merge sort divides into halves and combines them in sorted order. Dividing alone is not enough; correctness depends on valid preconditions and combining.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Search sorted [2,5,8,11,14,17,20,23] for 20.

Follow the reasoning

Each iteration narrows the candidate interval, and result is index 6.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For [1,4,7,10,13,16,19,22] trace middle indexes when target19. [6]

Question 2

Explain divide, solve and combine for merge sort on [7,2,9,1]. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare repeated halving with linear scan for sorted list of 1024 items, in terms of worst-case comparisons and preconditions. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Binary search will correctly find any target in any unsorted list.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.